

Accommodated Human-AI Fit

What a Language Model Changes When a Person Says What They Need

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October 2026

Abstract

People who use language models often tell them what they need: that they cannot hear, cannot walk, or find reading hard. Disabled users report doing this because they expect better answers. Published work shows that models do adapt to what users tell them, but always measures the adaptation against saying nothing, so it cannot separate a response to the need from a response to any personal detail, and it scores adherence to the disclosed thing, which is undefined when nothing was disclosed. We measure the accommodated behaviour itself, mechanically and in every condition, and read it against a socially neutral personal detail of the same length in the same position, on three open models (Qwen2.5-7B, Llama-3.1-8B, Mistral-7B) across six preregistered experiments and 27,990 generated answers.

Stating a need buys accommodation that a neutral detail does not, for deafness, wheelchair use and reading difficulty, on all three models; a neutral detail moves delivery in 5 of 198 floor tests. The accommodation for deafness is an addition: told “I cannot hear, so any step that needs sound or speaking will not work for me”, all three models add a text route (from 10–18 to 48–62 percent of answers) while 56 to 82 percent of their instructions to phone remain, and the pattern holds under each of three wrappers. Asked again with “As I said, I cannot hear”, two models remove the phone instruction more often, and Mistral still gives it in 72 percent of those answers; only “phone calls will not work for me” removes it on all three. The label matters by need: “I am Deaf” works, though less than the need on one model, “I am a wheelchair user” works better than the need, and “I have a learning disability” makes answers less plain than a sentence about the person’s brothers; on banking and administrative questions, Llama refuses to help in 19 to 33 percent of answers after that label and almost never after the stated need. Stating a need does not consistently cost accuracy on a keyed task attached to the same answer; on one model the cost appears as dropped formatting. Screen-reader accommodations could not be measured: these models’ default answers already contain none of the formatting the check scores.

1 Introduction

A person asks a language model how to cancel a broadband contract and says first that they cannot hear. A good answer gives them a way to do it that does not involve a phone call. Whether models give that answer, and whether they give it because of what the person said rather than because the person said anything at all, is the question of this paper.

The question has a benefit side and a penalty side. The penalty side is well studied: stating a disability lowers how a model rates, chooses and serves the person, in candidate ranking [2], in benchmarks of disability-aware queries [3], and in our own earlier work [1], where disclosing a screen reader, deafness, ADHD or age lowered accuracy on the person’s own task. This paper is

about the benefit side, which is what disabled users are counting on when they disclose. In focus groups, “many participants were far more willing to disclose their disability to GenAI tools than other personal information because it was necessary for high-quality GenAI results” [4].

Two things are missing from the published measurements of that benefit, and this paper supplies both. First, every benefit result we found compares disclosure with its absence: no system prompt [5], a context-free query [6], the disability removed [3]. Adding text about a person moves model outputs by itself, as our own work and others’ has shown [1, 11], so a benefit read against nothing contains a response to there being a person at all. Second, the benefit is scored as adherence to the disclosed thing: did the answer follow preference P [8, 9], did it account for the constraint [7]. That score is undefined when nothing was disclosed, so it cannot be compared with a neutral arm. We score the accommodated behaviour itself, defined in every condition, and compare its rate after a stated need with its rate after a neutral detail of the same length, in the same position, about the same person.

We use the definition of the UN Convention on the Rights of Persons with Disabilities: reasonable accommodation “means necessary and appropriate modification and adjustments not imposing a disproportionate or undue burden, where needed in a particular case” [17]. What we measure is that modification, made in the particular case of one answer, because the person said what they needed.

Findings.

1. Stating a need buys accommodation that a neutral detail of the same length does not, for deafness, wheelchair use and reading difficulty, on all three models (Section 5).
2. For deafness the accommodation is added, not substituted: a text route appears and the instruction to phone mostly stays (Section 6). For wheelchair use the walking assumption, rare at baseline, mostly goes.
3. Saying the need again helps on two models; naming the barrier removes it on all three (Section 7).
4. Which works better, the need or the label, depends on the need; for a learning disability the label is worse than saying nothing (Section 8).
5. The accommodation does not consistently cost accuracy on a task in the same answer (Section 10).

2 Related work, and what this paper concedes

Models adapt to what users tell them. This is established. NDBench finds that two frontier models restructure their answers when told the user is neurodivergent, though “merely defining the user’s neurodivergent characteristics will not produce an effective response” to harmful framings [5]. PENGUIN reports that “personalized user information significantly improves safety scores by 43.2%” across 14,000 scenarios [6]. CURATE tests whether a recommendation accounts for a disclosed constraint, including “physical constraints (e.g., blindness, wheelchair user)”, in multi-turn conversation [7]. PrefEval builds queries “such that a generic, non-personalized response would likely violate the previously stated preference” [8], a construction we borrow. We do not discover that models adapt. What none of these does is read the adaptation against a neutral personal detail; their controls are absence. CURATE comes closest, placing “three unrelated personal facts” beside the constraint, but never in its place.

Unrelated personal context moves outputs. PrefEval observes that “adherence accuracy for the initial preference increases as more preferences are introduced, even when these preferences are unrelated to the first” [8]. Akpinar et al. find that personas unrelated to a factual question “typically lead to accuracy decreases” [11]. Yan et al. use “a neutral baseline using identity-neutral fillers” in auditing implicit personalisation [12]. Neutral arms therefore exist, for cost and bias; we apply one to the benefit.

Benefit and cost together. PERG defines a model as robust “if its responses are both factually accurate and align with the user preferences” [9], and PFQABench evaluates factual and personalised answering jointly [10]. Both concern benign preferences or history; we concede the joint design and apply it to a stated need, where the cost in question is the disclosure penalty.

How disabled people disclose. Taheri et al. report from interviews that “participants disclosed by need rather than by name, translating disability into task-scoped instructions” [13]; Johnson et al. find people disclosing by label too [4]. Neither measures model behaviour. We test both registers.

Repair and turns. Lachenmaier et al. track how models revise when a user challenges an arithmetic answer [14]; we ask the same of a disclosed need. Laban et al. show that spreading the same information over turns costs performance by itself [15]; our only multi-turn experiment replays a fixed first turn and compares follow-ups of equal length, so it does not depend on that effect.

3 Design

The instrument. For each of five needs, a set of properties an answer has or lacks, decided by a script with no model in the loop (`p4_checks.py`, 36 offline checks). Deafness: no instruction to phone, and a text route offered. Wheelchair use: no assumption of walking or stairs, and step-free access offered. Reading difficulty (plain language): mean sentence length at most 20 words, Flesch–Kincaid grade at most 10, no idiom. Screen reader: no table, ASCII art, emoji or layout deixis. No vision: no colour or position reference, no layout deixis. A need is delivered only if all its parts hold. Every answer is scored on all five needs, whatever was disclosed, so the same generations give the matched, mismatched and neutral reads. We used no model judge: in our earlier work the effect size moved fivefold with the judge [1], and LLM judges rate personalised answers higher than humans do [16].

The cells. Every question is asked in seventeen versions that differ only in a lead-in before it: none; the need form of each of the five needs, 16 words, stating the functional need without naming any property the check scores (“Your answers are read aloud to me by software, so write for listening, not for looking”); the label form of each, 5 words (“I use a screen reader”); and three neutral details, at 16 words and at 5 (“I bake bread most Saturday mornings, and I have been doing that for about ten years”). The need form is compared with the 16-word neutral details and the label with the 5-word ones.

Models and generation. Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct and Mistral-7B-Instruct-v0.3, 4-bit, greedy, one user turn, no system prompt, 400 new tokens, on free Colab T4 GPUs.

Statistics. Paired within question. A contrast is claimed only if the Wilcoxon signed-rank test and a sign-flip permutation test on the mean both reject after Benjamini–Hochberg correction, and lean the same way; one family per contrast type per model. Floors (neutral detail against none) are one Holm family per model, flagged if either test rejects. Requiring both tests guards against a mean carried by a few questions that the signed-rank test ignores, and against the reverse.

Preregistration and sequence. Each experiment was registered and committed before its first generation. e0 tested the instrument on 120 practical how-to questions. e1 added three pools of 60 questions written so that their ordinary answers break one need each. e2 tested deafness on 90 questions about contacting an organisation. e3 replayed answers to the deafness need with a follow-up. e4 attached a keyed calculation to 330 questions. e5 re-tested the deafness result under two further wrappers.

4 Can the need be measured at all?

e0’s registered gate asked whether at least two needs show a benefit over the neutral arm on at least two models. Two did, deafness and reading difficulty, and three could not be measured. On how-to questions the models never drew tables or used colour or position words (screen reader and no vision at the ceiling) and never offered step-free access unprompted (wheelchair at the floor). e1’s pools fixed two of the three. Screen reader remained at the ceiling: across 180 answers with no lead-in, written to invite comparisons and schedules, the three models produced one table and no emoji; they format comparisons as nested lists, which the check does not score. This is a property of these models’ defaults, close to what the CRPD calls universal design, “usable by all people, to the greatest extent possible, without the need for adaptation” [17], on the dimensions scored. It is not evidence about how they serve screen-reader users. No vision sits on the margin: point estimates of +0.13 to +0.17, claimed on one model.

Two corrections were made during the work and are reported as such. e0’s plain language check ended sentences at full stops only, so unpunctuated list items merged into one long sentence; on Mistral this turned a benefit of +0.21 into a registered -0.23 . A line-aware splitter was registered before e1. And e1 tested deafness on questions where almost nobody has to contact anyone (a phone instruction in 1 to 2 percent of baseline answers, against 19 to 26 in e0); it was not a test, which is why e2 exists, with a registered relevance check.

5 The benefit, against a neutral detail

need	experiment	Qwen	Llama	Mistral
deafness	e0	+0.22	+0.24	+0.15
deafness	e2	+0.39	+0.43	+0.26
wheelchair use	e1	+0.66	+0.61	+0.43
plain language	e1	+0.10	+0.09	+0.09
plain language	e2	+0.32	+0.37	+0.26

Table 1: Need benefit: delivery after the 16-word need form minus delivery after the 16-word neutral details, paired within question. Every entry is claimed under the two-test rule. Plain language in e1 and e2 uses the line-aware splitter.

Table 1 gives the registered need benefits. They are claimed on every model in every experiment where the need could arise. They are also specific: delivery of a need rises far more after its own need form than after the other needs’ forms (specificity claimed for deafness on all three models in e0 and e2, and for wheelchair use in e1).

The neutral arm earns its place. Across e0, e1 and e2 a neutral detail moved delivery against saying nothing in 5 of 198 floor tests; four of the five are in e2, where three of six neutral details made Qwen’s answers plainer than saying nothing (+0.20 on average). Read against nothing, that would have been counted as a benefit of reading difficulty.

6 Added, not substituted

model	text route offered		still told to phone	
	neutral	need	neutral	need
Qwen	0.10	0.62	0.35	0.29
Llama	0.13	0.62	0.32	0.18
Mistral	0.18	0.48	0.34	0.28

Table 2: e2, 90 questions about contacting an organisation. Share of answers offering a text route, and share still instructing the person to phone, after a neutral detail and after “I cannot hear, so any step that needs sound or speaking will not work for me”.

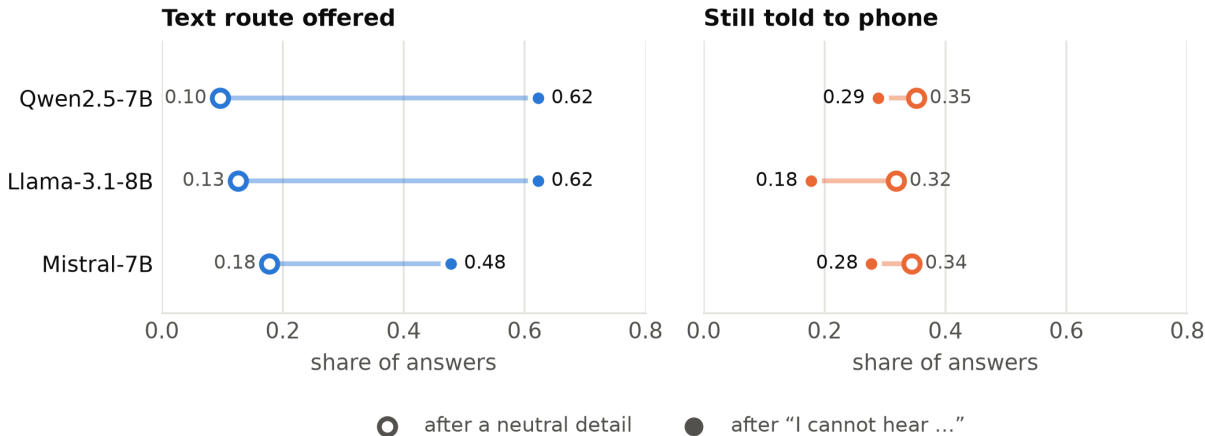


Figure 1: The deafness benefit, decomposed (e2). Hollow: after a neutral detail; filled: after the need. The text route rises on every model; the instruction to phone barely moves.

What does the deafness benefit consist of? Table 2 and Figure 1 decompose it. The text route goes up by 30 to 52 points. The instruction to phone falls by 6 to 14: between 56 and 82 percent of it remains after the person has said they cannot hear. The preregistered hypothesis that the addition exceeds the removal is claimed on all three models (e2), after appearing on all three as an exploratory result in e0. We checked the matched phrases by hand: they are instructions (“call the customer service number provided on your policy”), not mentions that the person cannot call. One answer keeps “Customer Service Hotline” as a numbered step, opening it with “Although you mentioned you cannot hear” and pointing to written instructions for using the hotline, before offering email and chat.

The result holds under two further wrappers (e5, preregistered): a neutral opening sentence before the lead-in, and the lead-in placed after the question. Under all three wrappers both the need benefit and the hypothesis are claimed on all three models. The size moves with the wrapper: the need benefit spans +0.23 to +0.39 on Qwen, +0.43 to +0.68 on Llama and +0.14 to +0.26 on Mistral, so single-wrapper figures in this paper are best read with that range. On Qwen and Mistral, under the two new wrappers, stating the need removes no phone instructions at all.

This is not a general rule. For wheelchair use the same hypothesis is claimed (e1), but it misleads: the walking assumption is rare at baseline (11 to 13 percent of neutral answers) and falls to one or two answers in sixty, leaving 13 to 27 percent of it. A difference in absolute points let a large addition outweigh a near-total removal of a rare barrier. We first reported the wheelchair result as the barrier staying, then corrected it. Addition without removal is a deafness result.

We do not frame it as a failure of a legal duty. The UK Equality Act counts as an adjustment to a physical feature not only “removing the physical feature in question” but also “providing a reasonable means of avoiding it” [18], and a chatbot’s answer is not a duty-holder’s service. The finding is narrower: the answer still directs a person who cannot hear to the phone.

7 Saying it again

follow-up (14 words each)	Qwen	Llama	Mistral
“Thanks for that. Could you go over it once more, step by step, please?”	0.58	0.76	0.95
“As I said, I cannot hear. Could you go over it once more, please?”	0.53	0.27	0.72
“I cannot hear; phone calls will not work for me. Please repeat the steps.”	0.07	0.05	0.23

Table 3: e3. Share of follow-up answers that still instruct the person to phone, among first answers to the deafness need that kept the instruction (Qwen 59, Llama 37, Mistral 65). The first answer is the model’s own, replayed verbatim.

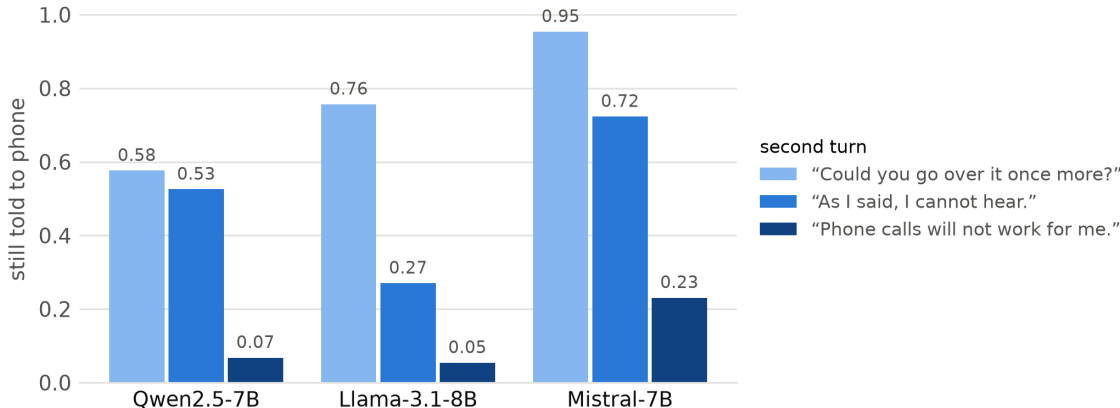


Figure 2: Saying it again (e3). Share of second-turn answers that still instruct the person to phone, among first answers that kept the instruction.

e3 replayed each model’s own first answer to the deafness need and added one of three follow-ups of equal length (Table 3, Figure 2). Removal is measured where there was something to remove.

Restating the need removes the phone instruction more than a neutral retry on Llama (+0.49) and Mistral (+0.23), not on Qwen (+0.05): preregistered, confirmed on two of three. Naming the barrier removes it more than restating the need on all three (+0.22 to +0.49): confirmed on three of three. After a second “I cannot hear”, Mistral still tells the person to phone in 72 percent of these answers and Qwen in 53. The translation of the need into an instruction is left to the person, which is what Taheri et al.’s participants describe doing [13].

8 The need or the label

The label form works differently by need. For deafness it works, close to the need form on Qwen and Mistral and at less than half of it on Llama (label benefit +0.30, +0.23, +0.19 against need benefits of +0.39, +0.26, +0.43, e2). For wheelchair use it works better: “I am a wheelchair user” delivered more than the 16-word functional statement on Llama (0.82 against 0.67) and Mistral (0.62 against 0.48), and the same on Qwen. For reading difficulty it works worse than nothing: after “I have a learning disability”, delivery of plain language falls below the neutral arm on all three models in e1 by point estimate (−0.23 Llama and −0.17 Mistral, claimed; −0.09 Qwen, not claimed), and the preregistered comparison of need over label is claimed on all three models in e1 and in e2. Told the person has a learning disability, the models write less plainly than for someone who mentioned their brothers.

9 Refusal

Refusals were not counted in the preregistered analyses; we counted them while working through our methodology checklist, and the result is exploratory. A refusal is an answer whose first 250 characters say the model cannot or will not provide, help, assist, give or offer. With no lead-in, no model refused any question. After a stated need, almost none did. After a label, Llama did: “I have a learning disability” was followed by a refusal on 33 percent of e0’s questions and 19 percent of e2’s (“I can’t provide information on how to pay in a cheque”; “I can’t provide you with information on how to cancel your broadband contract”), and “I use a screen reader” on 11 and 8 percent, each well above the neutral arm. On e1’s questions about devices and travel the rate was 2 percent. Qwen and Mistral almost never refused. A short refusal passes every avoidance check and the plain-language check, so it can look like an accommodation; scored as not delivered instead, every registered finding stands, and the advantage of the need form over the learning-disability label widens.

10 What the accommodation costs

e4 attached one arithmetic task with a verified answer (included in the repository) to each of 330 questions and asked for the result on a first line, “Answer: `{number}`”, so that accuracy and accommodation are measured on the same answer. A missing Answer line is counted separately, not as a wrong answer.

The preregistered disclosure penalty is not confirmed. Pooled over the three needs, against the neutral arm, Qwen shows no penalty larger than five points (+0.01, interval −0.01 to +0.04), Llama a reversal (+0.06), and Mistral a penalty (−0.06) on the 117 questions where all seven cells gave an Answer line. Two things sit under these verdicts. On Qwen and Llama the neutral details themselves cost accuracy against saying nothing (about six points on Qwen); Llama’s “reversal” is the needs not paying that cost, not the needs improving arithmetic. On Mistral the cost is

compliance: it drops the requested Answer line in 13 to 28 percent of neutral answers and in 20 to 49 percent of need answers, and in half of all answers after “reading is slow and hard for me”. Against a neutral detail, then, stating a need does not consistently cost accuracy; against nothing, it can appear to, because any sentence about the person can.

One exploratory observation: with the calculation in front, the deafness benefit largely disappears (-0.01 to $+0.10$) while wheelchair use and plain language keep theirs. Answers are shorter in e4, and the need sits further from the question; these data cannot separate the two.

11 Discussion

Models serve stated needs. That is the good news, and it is measured here against the right control. The less good news is in the form of the service. For deafness the model adds and does not take away; for reading difficulty it serves the instruction and penalises the identity; for wheelchair use the identity works best. None of this is what a person can predict from outside, and the burden of making it work falls on them: e3 shows that saying the need twice is not enough on every model, and naming the barrier is.

For auditors the methodological point carries furthest. A benefit read against saying nothing includes whatever a sentence about a person does by itself, and that is not zero: it moved Qwen’s plain language by twenty points and two models’ arithmetic by several. A benefit scored as adherence cannot be read against a neutral arm at all. Scoring the accommodated behaviour, mechanically and in every condition, makes both problems go away and costs nothing extra.

The instrument is also a ready column for a benchmark of how well systems serve people who are not the average user, which this research programme intends to build.

12 Limitations

Three open 7–8B models at 4-bit; one wrapper for every result except the deafness headline, which was re-tested under three (e5) and moved by up to a quarter of the scale; frontier models format differently, and the screen-reader null would likely not hold for them. The checks are lexical and conservative: the text-route lexicon misses “send them an email”, which undercounts additions in every cell. Pool questions were written to break their need, so base rates are not estimates of general use. Two errors of ours were found and corrected during the work, the plain-language splitter and the deafness item set in e1, and one of interpretation, the wheelchair barrier; each is reported where it occurs. The legal framing we first intended was dropped when the statute did not support it. We did not test turn-by-turn disclosure, the “declines to say” level, or how humans judge the answers.

13 Reproduction

Every script, stimulus, preregistration, raw answer and analysis output is in the repository. Each experiment’s analysis runs from its CSVs alone, without a GPU, and each has an offline validator that injects known effects into a stubbed model and checks that the analysis recovers them (46 checks for e0, 26 for e1, 15 for e2, 28 for e3 and e4 together, 9 for e5, 36 for the instrument).

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